EXP.NO: 5

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# DEVELOP A CUSTOMIZED PING COMMAND TO TEST SERVER CONNECTIVITY

**Aim:**

To develop a customized Ping commands to test Server Connectivity

**PURPOSE OF USING PING COMMAND:**

Ping Commands are used to

- Verify whether a Server or host is reachable
- Test the connectivity between two devices
- Measure the Round Trip Time (RTT) or latency.
- Check for packet loss during communication

**PURPOSE OF PING COMMAND:**

-> Ensure that communication with the destination Server is Successful

-> Evaluate the quality and reliability of a network connection

-> Detects delays in data transmission

-> Identify packet loss that may affect network performance.

Command: ping google.com

OUTPUT:

Pinging google.com [142.250.77.110] with 32 bytes of data:
Reply from 142.250.77.110: bytes=32 time=13ms TTL=119
Reply from 142.250.77.110: bytes=32 time=4ms TTL=119
Reply from 142.250.77.110: bytes=32 time=7ms TTL=119

Ping statistics for 142.250.77.110:

Packets: sent=4, Received=4, Lost=0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 13ms, Average = 7ms

Command: ping -n 3 google.com

OUTPUT:

Pinging google.com [142.250.77.110] with 32 bytes of data:
Reply from 142.250.77.110: bytes=32 time=20ms TTL=119
Reply from 142.250.77.110: bytes=32 time=25ms TTL=119
Reply from 142.250.77.110: bytes=32 time=25ms TTL=119

Ping statistics for 142.250.77.110:

Packets: Sent=3, Received=3, Lost=0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 20ms, Maximum = 25ms, Average = 23ms

Command: ping -l 1000 google.com

OUTPUT:

Pinging google.com [142.250.67.46] with 1000 bytes of data:
Request timed out.
Reply From 142.250.67.46 : bytes = 1000 time = 3ms TTL = 118
Reply from 142.250.67.46 : bytes = 1000 time = 8ms TTL = 118
Reply from 142.250.67.46 : bytes = 1000 time = 5ms TTL = 118

Ping statistics for 142.250.67.46:

Packets: Sent = 4, Received = 3, Lost = 1 (25% Loss),
Approximate round trip times in milli-seconds:

Minimum = 3ms, Maximum = 8ms, Average = 5ms

command: ping -w 3000 google.com

OUTPUT:

Pinging google.com [142.250.67.46] with 32 bytes of data:
Reply from 142.250.67.46 : bytes = 32 time = 3ms TTL = 118
Reply from 142.250.67.46 : bytes = 32 time = 5ms TTL = 118
Reply from 142.250.67.46 : bytes = 32 time = 9ms TTL = 118
Reply from 142.250.67.46 : bytes = 32. time = 12ms TTL = 118

Ping statistics for 142.250.67.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% Loss)
Approximate round trip times in milliseconds

Minimum = 3ms, Maximum = 12ms, Average = 7ms

## CONNECTION ESTABLISHMENT PROCEDURE

1. Open the command prompt or terminal
2. Execute the required customized ping command
3. Observe the transmitted and received packet information
4. Record the response time and packet loss
5. Analyze the network performance based on the obtained results.

## CUSTOMIZED PING COMMANDS.

| Command | Purpose |
|---|---|
| ping google.com | check connectivity with the server |
| ping -n 10 google.com | Sends 10 ICMP Echo Requests. |
| Ping -t google.com | continuously sends ping request until stopped |
| ping -l 1000 google.com | send packets with a payload size of 1000 bytes |
| ping -w 3000 google.com | waits upto 3000ms for each reply |

## RESULT:

Thus customized Ping commands are developed and the server connectivity is tested Successfully